

Average Precision Under Random Ranking: Exact Expectation and Distribution

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Abstract

We derive the exact finite-sample expectation and probability distribution of Average Precision (AP) under uniformly random ranking, correcting the common approximation that equates expected AP with prevalence. The expectation has a harmonic-number closed form, while the full discrete law is a finite count over relevant-rank subsets, with an equivalent coefficient-generating formula. Our expectation result shows that $\mathbb{E}[\text{AP}] = p + O(\log L/L)$ where p is the prevalence and L the list length, revealing a persistent positive bias that affects statistical significance testing.

1. Introduction

Average Precision (AP) is a fundamental metric in information retrieval, machine learning, and ranking evaluation [Robertson 2008]. It summarizes the precision-recall trade-off into a single scalar and corresponds to the area under the uninterpolated precision-recall curve. Understanding its behavior under the null hypothesis of random ranking is essential for:

1. Statistical significance testing of ranking algorithms
2. Establishing baseline performance expectations
3. Detecting non-random patterns in ranked outputs

While it is commonly assumed that the expected AP under random ranking equals the prevalence (proportion of relevant items), Bestgen (2015) demonstrated that this is only an approximation. For finite samples—particularly common in specialized domains with limited labeled data—the exact expected value differs substantially from prevalence.

1.1 Contributions

This paper provides a comprehensive treatment of expected AP under random ranking:

1. **Closed-form Derivation:** We present a complete proof of the exact formula using exchangeability arguments (Theorem 1)
2. **Unified Presentation:** We provide an alternative derivation using harmonic numbers and connect it with Bestgen’s (2015) hypergeometric algorithm, showing their equivalence

3. **Asymptotic Analysis:** We characterize the precise $O(\log L/L)$ convergence rate with explicit constants
4. **Full Distribution:** We give the exact finite PMF, including the multiplicities caused by AP collisions
5. **Practical Implications:** We quantify the impact on statistical testing and provide implementation guidance

2. Problem Formulation

2.1 Notation and Definitions

Consider a ranking of L items with binary relevance labels $y_1, \dots, y_L \in \{0, 1\}$, where:

- M items are relevant ($\sum_{i=1}^L y_i = M$)
- $N = L - M$ items are non-relevant
- $p = M/L$ denotes the prevalence

Definition 1 (Average Precision). The Average Precision of a ranking is:

$$\text{AP} = \frac{1}{M} \sum_{k=1}^L \text{Prec}@k \cdot y_k = \frac{1}{M} \sum_{k: y_k=1} \text{Prec}@k$$

where $\text{Prec}@k = \frac{1}{k} \sum_{j=1}^k y_j$ is the precision at rank k .

Definition 2 (Random Ranking). A uniformly random ranking is a permutation of the L items chosen uniformly from all $L!$ possible permutations.

2.2 The Approximation Gap

The naive approximation $\mathbb{E}[\text{AP}] \approx p$ assumes that under random ranking, relevant items are uniformly distributed, leading to a constant precision-recall curve at height p . However, this ignores the discrete nature of finite samples and the dependency structure in the precision calculations.

3. Main Results

3.1 Exact Expected Value

Theorem 1 (Closed-form Expected AP). Under uniformly random ranking of $L = M + N$ items with M relevant items, the expected Average Precision is:

$$\mathbb{E}[\text{AP}] = \frac{1}{L} \left[\frac{M-1}{L-1} (L - H_L) + H_L \right]$$

where $H_L = \sum_{k=1}^L \frac{1}{k}$ is the L -th harmonic number.

Proof. We exploit the exchangeability of items under random ranking. Consider a uniformly chosen relevant item at rank R , where $P(R = r) = \frac{1}{L}$ for $r \in \{1, \dots, L\}$.

Given $R = r$, let X denote the number of other relevant items among the first $r - 1$ positions. By the hypergeometric distribution:

$$X \mid R = r \sim \text{Hypergeometric}(L - 1, M - 1, r - 1)$$

$$\text{with } \mathbb{E}[X \mid R = r] = \frac{(r-1)(M-1)}{L-1}.$$

The precision at rank r is:

$$\text{Prec}@r = \frac{X + 1}{r}$$

Therefore:

$$\mathbb{E}[\text{Prec}@r \mid R = r] = \frac{1}{r} \left(\frac{(r-1)(M-1)}{L-1} + 1 \right) = \frac{(r-1)(M-1)}{(L-1)r} + \frac{1}{r}$$

By the exchangeability argument, each relevant item contributes equally to the expected AP, so:

$$\mathbb{E}[\text{AP}] = \mathbb{E}[\text{Prec}@R] = \frac{1}{L} \sum_{r=1}^L \left(\frac{(r-1)(M-1)}{(L-1)r} + \frac{1}{r} \right)$$

Using the identities:

- $\sum_{r=1}^L \frac{r-1}{r} = \sum_{r=1}^L \left(1 - \frac{1}{r}\right) = L - H_L$
- $\sum_{r=1}^L \frac{1}{r} = H_L$

We obtain:

$$\mathbb{E}[\text{AP}] = \frac{1}{L} \left[\frac{M-1}{L-1} (L - H_L) + H_L \right] \quad \square$$

3.2 Alternative Forms

Corollary 1 (Prevalence-plus-correction form). *The expected AP can be expressed as:*

$$\mathbb{E}[\text{AP}] = p + \frac{N(H_L - 1)}{L(L-1)}$$

where the second term represents the finite-sample correction.

Proof. Algebraic manipulation of Theorem 1 yields:

$$\mathbb{E}[\text{AP}] = \frac{M}{L} + \frac{N}{L(L-1)} (H_L - 1) = p + (1-p) \frac{H_L - 1}{L-1} \quad \square$$

3.3 Algorithmic Derivation

Theorem 2 (Bestgen 2015). *The expected AP can be computed algorithmically as:*

$$\mathbb{E}[\text{AP}] = \frac{1}{M} \sum_{i=1}^M \sum_{n=i}^{N+i} P_{\text{hyper}}(i; L, M, n) \cdot \left(\frac{i}{n}\right)^2$$

where $P_{\text{hyper}}(i; L, M, n)$ is the hypergeometric probability mass function.

Proof Sketch. For each i -th relevant item:

1. It can appear at ranks $n \in \{i, i+1, \dots, N+i\}$
2. The probability of exactly i successes in the first n draws is hypergeometric
3. Given i successes in n draws, the probability the i -th occurs at position n is i/n
4. The precision at rank n with i relevant items is i/n

The double summation aggregates these contributions. $\square$

Remark. Bestgen’s algorithmic approach, while computationally more intensive ($O(ML)$ operations), provides insight into the probabilistic structure and serves as independent validation of Theorem 1. We include it here to demonstrate the equivalence of the two methods.

3.4 Exact AP Distribution

Let $1 \leq R_1 < \dots < R_M \leq L$ be the ordered ranks of the relevant items. A uniform item permutation induces a uniform choice among the $\binom{L}{M}$ possible relevant-rank sets: every set has exactly $M!(L-M)!$ permutation preimages. At rank R_j , exactly j relevant items have appeared, so

$$A = \text{AP} = \frac{1}{M} \sum_{j=1}^M \frac{j}{R_j}.$$

Theorem 3 (Exact AP PMF). *For $1 \leq M \leq L$ and any rational a ,*

$$\Pr(A = a) = \frac{1}{\binom{L}{M}} \sum_{1 \leq r_1 < \dots < r_M \leq L} \mathbf{1} \left\{ a = \frac{1}{M} \sum_{j=1}^M \frac{j}{r_j} \right\}.$$

Equivalently, the AP law is the finite mixture

$$\mathcal{L}(A) = \frac{1}{\binom{L}{M}} \sum_{1 \leq r_1 < \dots < r_M \leq L} \delta_{\frac{1}{M} \sum_{j=1}^M j/r_j}.$$

This formula retains multiplicity because the map from rank sets to AP values is not injective. For example, when $L = 6$ and $M = 2$, rank sets $(2, 6)$ and $(3, 4)$ both give $A = 5/12$, so $\Pr(A = 5/12) = 2/15$.

A coefficient form makes the same multiplicities explicit. Let $D = L!$, define

$$K(r_1, \dots, r_M) = \sum_{j=1}^M j \frac{D}{r_j},$$

and let

$$P_{L,M}(z) = \sum_{1 \leq r_1 < \dots < r_M \leq L} z^{K(r_1, \dots, r_M)}.$$

Then every AP value lies on the rational grid $k/(MD)$ and

$$\Pr\left(A = \frac{k}{MD}\right) = \frac{[z^k]P_{L,M}(z)}{\binom{L}{M}}, \quad P_{L,M}(1) = \binom{L}{M}.$$

For $M = 0$, the repository's convention gives the point mass $A = 0$. For $M = L$, the law is the point mass $A = 1$. The Lean theorem `uniformAPMass_closed_form_explicit` proves the subset-count formula directly from the original uniform-permutation definition.

4. Asymptotic Analysis

4.1 Convergence Rate

Theorem 4 (Asymptotic Behavior). *As $L \rightarrow \infty$ with prevalence p held fixed:*

$$\mathbb{E}[\text{AP}] = p + \frac{(1-p) \log L}{L} + O\left(\frac{1}{L}\right)$$

Proof. Using the well-known asymptotic expansion of the harmonic number [see, e.g., Graham et al. 1994], $H_L = \log L + \gamma + \frac{1}{2L} + O(L^{-2})$ where $\gamma \approx 0.5772$ is the Euler-Mascheroni constant:

$$\mathbb{E}[\text{AP}] - p = \frac{N(H_L - 1)}{L(L-1)} = \frac{(1-p)(\log L + \gamma - 1)}{L-1} + O(L^{-2})$$

The dominant term is $\frac{(1-p) \log L}{L}$. $\square$

4.2 Practical Implications

Corollary 2. *The relative error $(\mathbb{E}[\text{AP}] - p)/p$ is:*

$$\frac{\mathbb{E}[\text{AP}] - p}{p} = \frac{1-p}{p} \cdot \frac{H_L - 1}{L-1} \approx \frac{1-p}{p} \cdot \frac{\log L}{L}$$

This shows the bias is most pronounced when:

1. Prevalence is low (small p)
2. Sample size is small (small L)

Example. For $p = 0.1$:

- $L = 100$: relative error $\approx 38\%$
- $L = 1000$: relative error $\approx 6\%$
- $L = 10000$: relative error $\approx 0.8\%$

5. Statistical Testing Implications

5.1 Hypothesis Testing

For testing whether an observed AP differs significantly from random:

Null Hypothesis: H_0 : Ranking is uniformly random **Test Statistic:** Observed AP

Under H_0 , the expected value is **not** the prevalence but rather given by Theorem 1. Using the incorrect null expectation (prevalence) leads to:

1. **Type I Error Inflation:** Falsely rejecting H_0 when AP exceeds prevalence but not $\mathbb{E}[\text{AP}]$
2. **Power Reduction:** Requiring larger effect sizes to achieve significance

5.2 Exact Variance and Tail Probabilities

The full PMF in Theorem 3 already determines the exact finite-sample variance:

$$\text{Var}(A) = \sum_a a^2 \Pr(A = a) - \mathbb{E}[A]^2.$$

It also gives exact upper-tail probabilities for an observed value a_{obs} :

$$\Pr(A \geq a_{\text{obs}}) = \sum_{a \geq a_{\text{obs}}} \Pr(A = a).$$

Thus neither quantity is mathematically unknown. The remaining problems are to simplify the variance into a compact analytic expression comparable to Theorem 1, and to compute tail probabilities without enumerating up to $\binom{L}{M}$ rank sets.

6. Implementation Notes

6.1 Computational Considerations

1. **Harmonic Numbers:** For $L \leq 10^6$, direct summation is efficient
2. **For larger L :** Use approximation $H_L \approx \log L + \gamma + \frac{1}{2L}$
3. **Hypergeometric Method:** Suitable for validation but computationally intensive for large L

6.2 Code Availability

Python implementations of both methods are available at: [repository URL]

7. Related Work

- **Robertson (2008)**: Introduced alternative AP formulations but assumed asymptotic behavior
- **Bestgen (2015)**: First formally identified that the common prevalence approximation is incorrect; provided the exact algorithmic approach using hypergeometric distributions
- **Yilmaz et al. (2008)**: Studied AP variance but under different assumptions

8. Conclusion

We have provided exact finite-sample formulas for both the expectation and full discrete distribution of Average Precision under random ranking. The harmonic expectation reveals the bias in the prevalence approximation, while the rank-subset PMF gives exact atom and tail probabilities.

Future Directions

The first and third items concern analytic simplicity and computational scale, not the existence of exact finite answers.

1. Derive a compact analytic formula for the second moment and variance
2. Extend the analysis to graded-relevance metrics such as NDCG and ERR
3. Develop scalable exact algorithms or controlled approximations for tail probabilities at large L and M

References

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Appendix A: Proof of Harmonic Identity

Lemma A.1. $\sum_{r=1}^L \frac{r-1}{r} = L - H_L$

Proof.

$$\sum_{r=1}^L \frac{r-1}{r} = \sum_{r=1}^L \left(1 - \frac{1}{r}\right) = L - \sum_{r=1}^L \frac{1}{r} = L - H_L \quad \square$$

Appendix B: Numerical Validation

We validate our formulas through three independent methods:

1. **Closed-form formula** (Theorem 1): Direct calculation using harmonic numbers
2. **Hypergeometric algorithm** (Theorem 2): Iterative computation following Bestgen (2015)
3. **Monte Carlo simulation**: Empirical estimation via random permutations

All three methods agree to machine precision for the exact methods (difference $< 10^{-15}$) and to sampling error for Monte Carlo ($< 10^{-4}$ with 10,000 trials). We tested configurations ranging from $L = 5$ to $L = 50,000$ with various prevalence levels.

The accompanying interactive notebook (`expected_ap.py`) provides:

- Complete numerical comparisons across all test cases
- Reproduction of Bestgen's Table 2
- Performance benchmarks comparing computational efficiency
- Visualization of convergence behavior

These empirical results confirm both the theoretical equivalence of the harmonic and hypergeometric approaches and the practical accuracy of the closed-form formula.